

Exercise 3: Creating Export Policies and Rules for Volumes

In this exercise, you review the default export policy and create a rule. Later, you create a new export policy and create a rule for it. In the end, you again try to mount the SVM namespace and verify the mounted file system information.

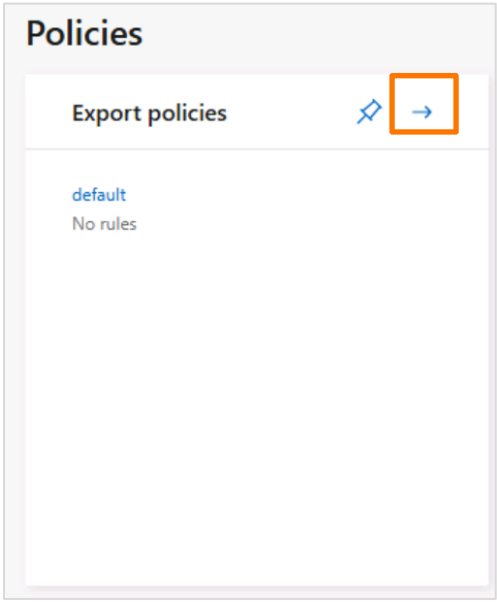
Objectives

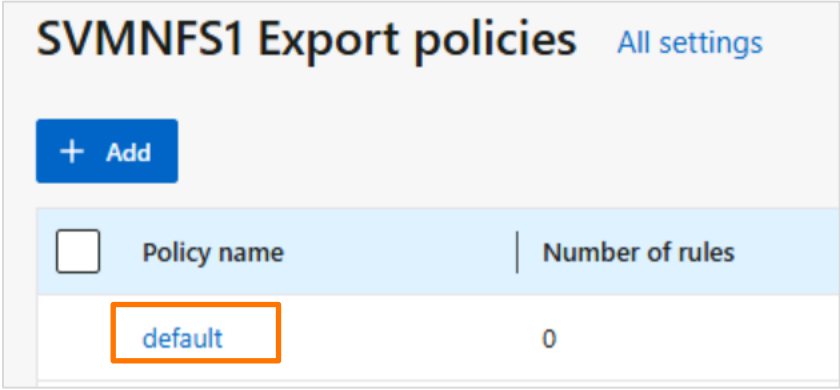
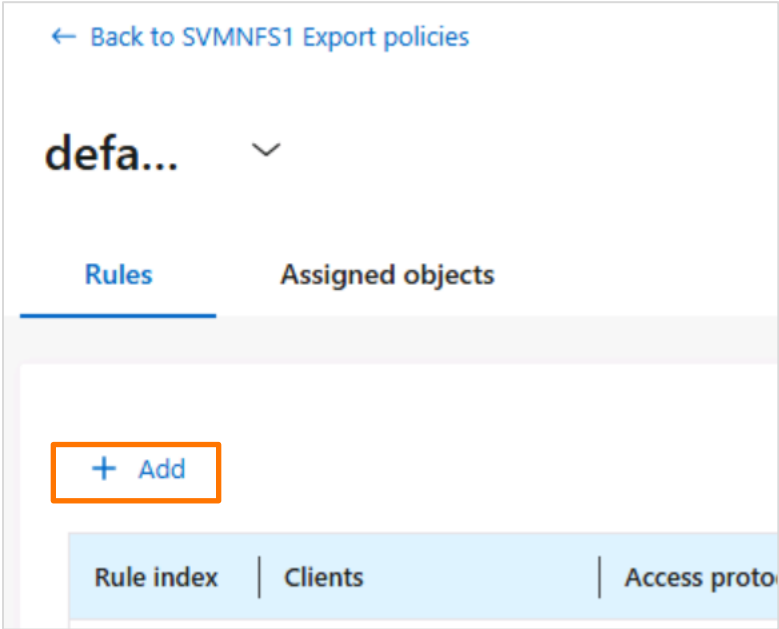
This exercise focuses on enabling you to do the following:

- Review the default export policy and create a rule
- Create a new export policy and create a rule for it
- Mount the SVM namespace

Task 1: Review the Default Export Policy and Create a Rule

In this task, you review the default export policy and create a rule.

Step	Action
1-1	From the ONTAP System Manager menu, select Cluster > Storage VMs .
1-2	Select SVMNFS1 and then click Settings .
1-3	In the Policies section, click the arrow that corresponds to Export policies. 

Step	Action
1-4	Click default. 
1-5	On the Rules tab, click + Add. 

Step	Action																																
1-6	In the New rule dialog box, specify the following details: - Client specification: 0.0.0.0/0 - Access protocols: - NFS: <select> - Read-only access: <select> - Read/Write access: <select> - Superuser access: <select> New rule Client specification 0.0.0.0/0 Access protocols ☐ SMB/CIFS ☐ FlexCache ☒ NFS ☒ NFSv3 ☒ NFSv4 Access details <table><thead><tr><th>Type</th><th>Read-only access</th><th>Read/write access</th><th>Superuser access</th></tr></thead><tbody><tr><td>All</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>All (as anonymous user) ⓘ</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>UNIX</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>Kerberos 5</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>Kerberos 5i</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>Kerberos 5p</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>NTLM</td><td>☒</td><td>☒</td><td>☒</td></tr></tbody></table> Cancel Save	Type	Read-only access	Read/write access	Superuser access	All	☒	☒	☒	All (as anonymous user) ⓘ	☐	☐	☐	UNIX	☒	☒	☒	Kerberos 5	☒	☒	☒	Kerberos 5i	☒	☒	☒	Kerberos 5p	☒	☒	☒	NTLM	☒	☒	☒
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NTLM	☒	☒	☒																														
1-7	Click Save .																																

Step

Action

1-8

Review the export policy rule.

defa... ▾

Rules

Assigned objects

+ Add

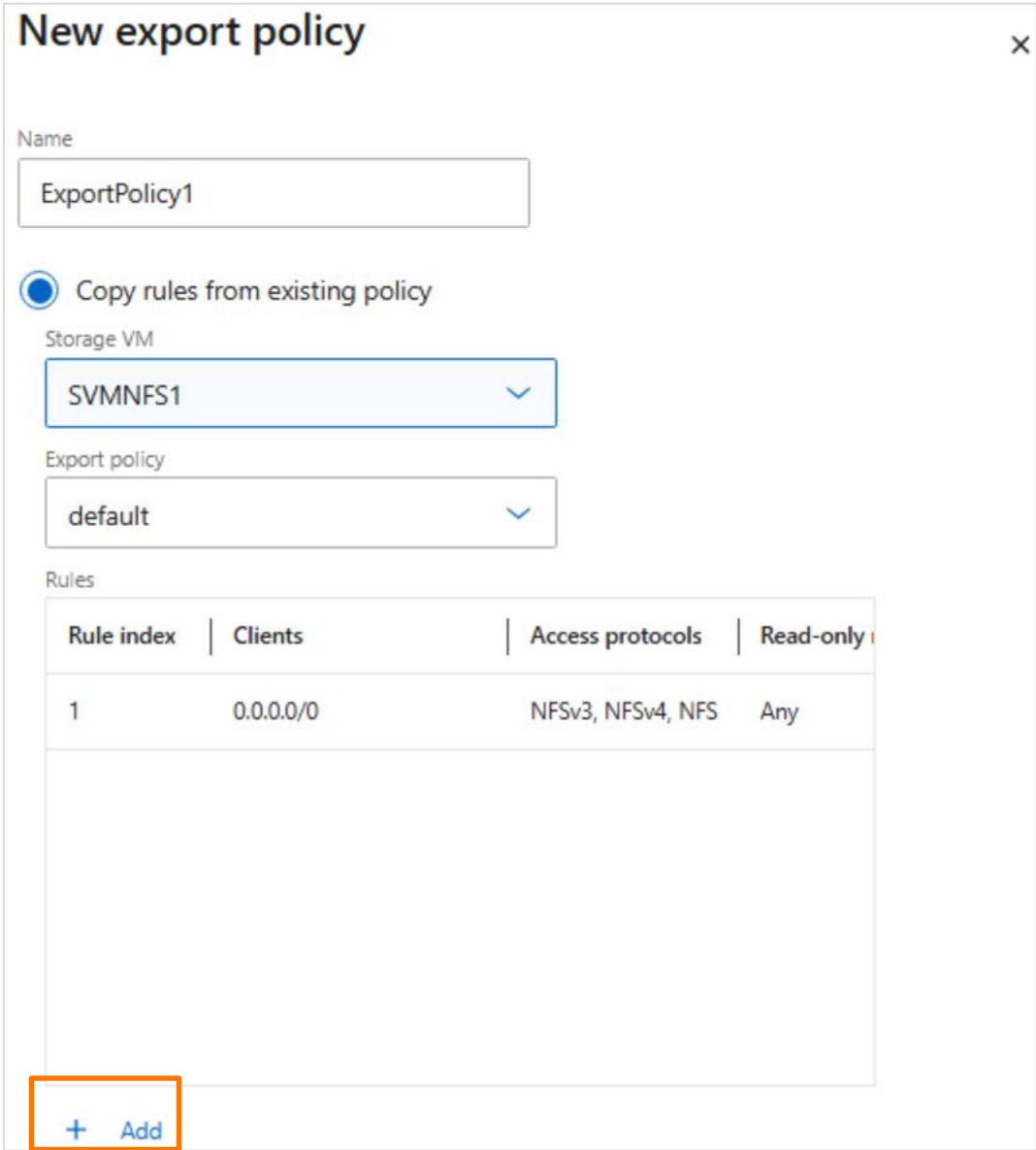
Rule index	Clients	Access protocols	Read-only rule	Read/write rule	SuperUser access	Anonymous user
1	0.0.0.0/0	NFSv3, NFSv4, NFS	Any	Any	Any	65534

Task 2: Create a New Export Policy and Create a Rule

In this task, you create an export policy and create a rule.

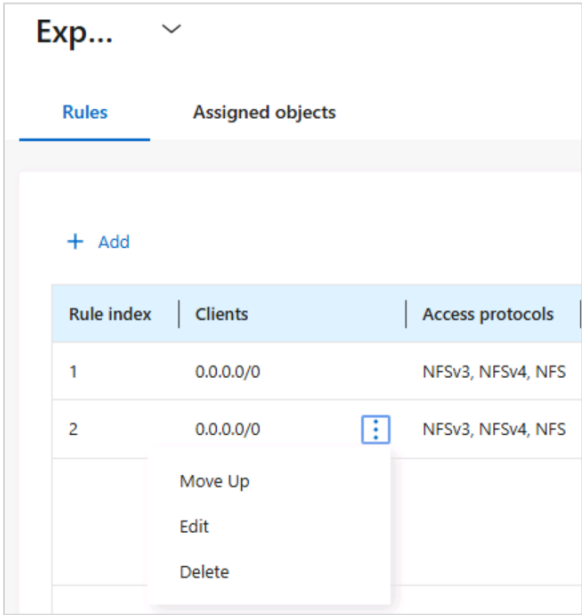
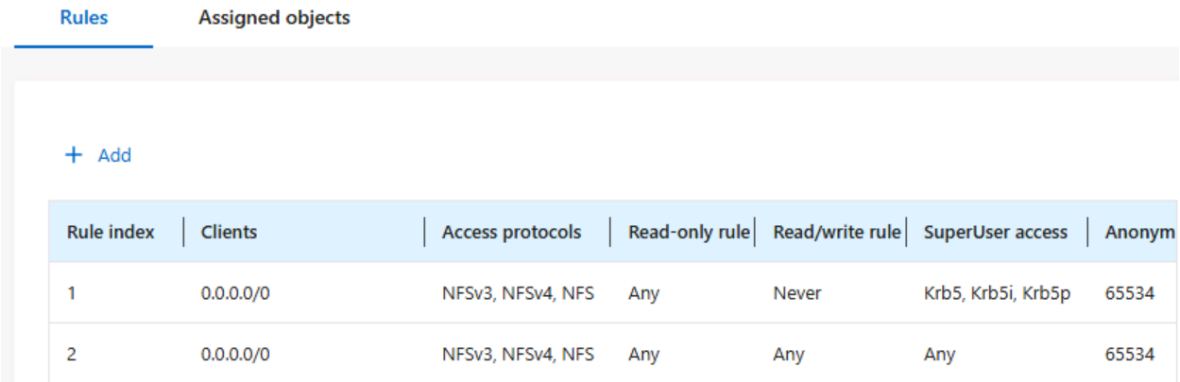
Step	Action						
2-1	Click Back to SVMNFS1 Export policies. ← Back to SVMNFS1 Export policies default ▼ Rules Assigned objects + Add <table><tr><th>Rule index</th><th>Clients</th><th>Access protocols</th></tr><tr><td>1</td><td>0.0.0.0/0</td><td>NFSv3, NFSv4, NFS</td></tr></table>	Rule index	Clients	Access protocols	1	0.0.0.0/0	NFSv3, NFSv4, NFS
Rule index	Clients	Access protocols					
1	0.0.0.0/0	NFSv3, NFSv4, NFS					
2-2	Click + Add to add an export policy. SVMNFS1 Export policies All settings + Add <table><tr><th>☐</th><th>Policy name</th><th>Number of rules</th></tr><tr><td></td><td>default</td><td>1</td></tr></table>	☐	Policy name	Number of rules		default	1
☐	Policy name	Number of rules					
	default	1					

Step	Action								
2-3	In the New export policy dialog box, specify the following details: • Name: ExportPolicy1 • Copy rules from existing policy: <select> • Storage VM: SVMNFS1 • Export policy: default New export policy Name ExportPolicy1 Copy rules from existing policy Storage VM SVMNFS1 Export policy default Rules <table><tr><th>Rule index</th><th>Clients</th><th>Access protocols</th><th>Read-only</th></tr><tr><td>1</td><td>0.0.0.0/0</td><td>NFSv3, NFSv4, NFS</td><td>Any</td></tr></table> + Add	Rule index	Clients	Access protocols	Read-only	1	0.0.0.0/0	NFSv3, NFSv4, NFS	Any
Rule index	Clients	Access protocols	Read-only						
1	0.0.0.0/0	NFSv3, NFSv4, NFS	Any						

Step	Action
2-4	Click + Add to add a new rule. 

Step	Action																																
2-5	In the New rule dialog box, specify the following details: - Client specification: 0.0.0.0/0 - Access protocols: - NFS: <select> - Read-only access: <select> - Superuser access (for Kerberos 5, Kerberos 5i, and Kerberos 5p): <select> New rule Client specification 0.0.0.0/0 Access protocols ☐ SMB/CIFS ☐ FlexCache ☒ NFS ☒ NFSv3 ☒ NFSv4 Access details <table><thead><tr><th>Type</th><th>Read-only access</th><th>Read/write access</th><th>Superuser access</th></tr></thead><tbody><tr><td>All</td><td>☒</td><td>☐</td><td>☐</td></tr><tr><td>All (as anonymous user) ⓘ</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>UNIX</td><td>☒</td><td>☐</td><td>☐</td></tr><tr><td>Kerberos 5</td><td>☒</td><td>☐</td><td>☒</td></tr><tr><td>Kerberos 5i</td><td>☒</td><td>☐</td><td>☒</td></tr><tr><td>Kerberos 5p</td><td>☒</td><td>☐</td><td>☒</td></tr><tr><td>NTLM</td><td>☒</td><td>☐</td><td>☐</td></tr></tbody></table> Cancel Save	Type	Read-only access	Read/write access	Superuser access	All	☒	☐	☐	All (as anonymous user) ⓘ	☐	☐	☐	UNIX	☒	☐	☐	Kerberos 5	☒	☐	☒	Kerberos 5i	☒	☐	☒	Kerberos 5p	☒	☐	☒	NTLM	☒	☐	☐
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2-6	Click Save twice.																																

Step	Action																					
2-7	Review the export policies and note that ExportPolicy1 has two rules. SVMNFS1 Export policies <a>All settings + Add <table><tr><th>☐</th><th>Policy name</th><th>Number of rules</th></tr><tr><td></td><td>default</td><td>1</td></tr><tr><td></td><td>ExportPolicy1</td><td>2</td></tr></table>	☐	Policy name	Number of rules		default	1		ExportPolicy1	2												
☐	Policy name	Number of rules																				
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	ExportPolicy1	2																				
2-8	Click ExportPolicy1 .																					
2-9	Review the Read/write rule and SuperUser access columns and notice the difference in the two rules. ExportPolicy1 ▼ Rules Assigned objects + Add <table><tr><th>Rule index</th><th>Clients</th><th>Access protocols</th><th>Read-only rule</th><th>Read/write rule</th><th>SuperUser access</th><th>Anonymous user</th></tr><tr><td>1</td><td>0.0.0.0/0</td><td>NFSv3, NFSv4, NFS</td><td>Any</td><td>Any</td><td>Any</td><td>65534</td></tr><tr><td>2</td><td>0.0.0.0/0</td><td>NFSv3, NFSv4, NFS</td><td>Any</td><td>Never</td><td>Krb5, Krb5i, Krb5p</td><td>65534</td></tr></table>	Rule index	Clients	Access protocols	Read-only rule	Read/write rule	SuperUser access	Anonymous user	1	0.0.0.0/0	NFSv3, NFSv4, NFS	Any	Any	Any	65534	2	0.0.0.0/0	NFSv3, NFSv4, NFS	Any	Never	Krb5, Krb5i, Krb5p	65534
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Step	Action
2-10	Click the ellipsis (three vertical dots) that correspond to Clients for Rule index 2 and select Move Up. 
2-11	 You see that rule index order has changed. 

Task 3: Mount the Storage VM Namespace Again

In this task, you mount the SVM namespace again.

Step	Action
3-1	In the Ubuntu client CLI, re-run the showmount -e command to display information about the junction paths. <pre>showmount -e 192.168.0.52</pre> <pre>root@ubuntu:/mnt# showmount -e 192.168.0.52</pre> Export list for 192.168.0.52: <pre>/ (everyone) /Vol1 (everyone) /Vol2 (everyone) root@ubuntu:/mnt#</pre>
3-2	Mount the storage VM namespace at the mount folder SVMNFS1 that you created in Exercise 2. <pre>mount -o nfsvers=3 192.168.0.52:/ /mnt/SVMNFS1</pre>
3-3	 The namespace is mounted successfully without issues this time because a rule was added to the default export policy, allowing access from the Linux client.

End of exercise